(4) (b)

Weighted least Squares minimizes the following objective error functions

$$E(w) = \frac{1}{2} \sum_{n=1}^{N} r_n \left(t_n - W^T(\phi(x_n))\right)^2$$

with $W_{new} = W_{old} - \theta_n \left(t_n - W^T \phi(x_n)\right) . x_n$

where θ_n = weight associated

t_n = value of nth data point

w = parameter to estimate

Newton or Raphson's method, tries to minimize the error function by iteratively updating weights using second order derivative with update equation as

$$W_{new} = W_{old} - \left(\nabla_w^2 L(w)\right)^{-1} \left(\nabla_w L(w)\right) \rightarrow (2)$$

eqn (2) is similar to eqn (1) as $\nabla_w^2 L(w)^{-1}$ i.e heissian^{-1} in Newton Raphson's method, corresponds to weight matrix in WLS.

$\nabla_W L(w)$ = gradient in Newton Raphson's method, corresponds to error term, i.e residual in WLS.

(ii) ~~Due to this~~ Newton Raphson's method iteratively reweighs data points by updating the weight vector in each iteration. Due to this re-weighing of data points, it is also called iterative least squares method.